

Practical 11 - Estimating causal effects using observational data 2

Survival Analysis

London School of Hygiene & Tropical Medicine

SOLUTIONS

Datasets required: `rotterdam_p11`, `rotterdam_p10`

R packages required: `survival`, `pec`, `adjustedCurves`, `riskRegression`

```
library(survival)
library(pec)
library(adjustedCurves)
library(riskRegression)
```

Introduction

In this practical we will continue to use the `rotterdam` data used in the lectures and in Practical 10. In Part A we will use a modified version of the data (`rotterdam_p11.csv`) in which some individuals change their hormone therapy use over time. In Part B we will return to using the data from Practical 10 (`rotterdam_p10.csv`), in which treatment status is fixed over time, and in which we will focus on competing risks outcomes.

Aims

The aims of this session are:

- Estimate conditional and marginal survival probabilities at a specific time under different treatment options, when observed treatment status changes over time.

- Estimate causal effects of a treatment in terms of marginal risk differences, risk ratios and restricted mean survival times, when observed treatment status changes over time.
- Estimate marginal survival curves under different treatment options, when observed treatment status changes over time.
- Estimate causal effects of treatments on competing events.

Part A: Changes in treatment status over time

We will use a modified version of the data, called `rotterdam_p11`. In this data some individuals not on hormone therapy at time 0 switch onto it at a later time. As in the previous practical the focus is on the composite event of disease recurrence or death from any cause, and on follow-up to 5 years. Administrative censoring at 5 years has already been applied. Variables used in this practical are summarised in the table below. We will also use the same set of adjustment variables as used in Practical 10 (`age`, `meno`, `size`, `grade`, `enodes`, `pgr`, `er`, `year_cat`, `chemo`).

This is not an official version of the data - we created it to illustrate the methods.

Variable name	Description
<code>pid</code>	Patient identifier
<code>hormon</code>	Ever used hormonal treatment: 0=no, 1=yes
<code>t_hormon</code>	Time dependent hormone treatment use: 0=no, 1=yes
<code>outcome</code>	Indicator of composite outcome: 0=censored, 1=event
<code>start_time</code>	Start time for this time interval
<code>stop_time</code>	Stop time for this time interval. Time of censoring or the composite event.
<code>rownum</code>	Row number for a given individual (1 or 2)
<code>nrow</code>	Ttal number of rows for a given individual (1 or 2)

Load the data:

```
rott2<-read.table("datasets/rotterdam_p11.csv",sep=";",header=T)
```

Question 1

- Take a look at the format of the data. How many rows of data are there and how many individuals?
- How many people switch onto hormone therapy over time (i.e. from `hormon=0` in their row 1 to `hormon=1` in their row 2)?

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```
#There are 2982 unique individuals (as before), but now 3246 rows of data
dim(rott2)
```

```
[1] 3246    17
```

```
length(unique(rott2$pid))
```

```
[1] 2982
```

```
#Number of people who switch from hormon=0 to hormon=1: 264
table(rott2$hormon[rott2$rownum==1 & rott2$nrow==2],
      rott2$hormon[rott2$rownum==2 & rott2$nrow==2])
```

```
    1
0 264
```

There are 2982 individuals in the data, but 3246 rows. There are 264 people who switch from `hormon=0` to `hormon=1` during their follow-up, and these individuals have two rows of data.

Question 2

In this question we will use the artificial censoring approach to handle the treatment switching and to estimate conditional survival probabilities under the treatment strategies of: (i) Start hormone therapy directly following surgery (and continue), (ii) Do not start hormone therapy directly following surgery or for the duration of follow-up.

Use the code below to implement the following:

- Modify the data so that people who switch from `hormon=0` to `hormon=1` are censored at the time of treatment switching. How many rows of data are there now?
- Fit a Cox model with hormone therapy as your main exposure, and conditioning on all the same set of variables as in the previous practical (`age`, `meno`, `size`, `grade`, `enodes`, `pgr`, `er`, `year_cat`, `chemo`).
- Interpret the hazard ratio estimate for the hormone therapy variable.

```
#data set with censoring at initiation of hormone therapy
#for people who switch from hormon=0 to hormon=1
roott2.cens<-roott2[(roott2$hormon==1 & roott2$rownum==1) |
                    (roott2$hormon==0),]

#Fit Cox model to the modified data
cox.adj<-coxph(Surv(stop_time,outcome)~hormon+age+meno+size+
               as.factor(grade)+enodes+pgr+er+year_cat+chemo,
               data=roott2.cens,x=TRUE)
```

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```
#data set with censoring at initiation of hormone therapy
#for people who switch from hormon=0 to hormon=1
roott2.cens<-roott2[(roott2$hormon==1 & roott2$rownum==1) |
                    (roott2$hormon==0),]

#We now have 2982 rows, as people with 2 rows of data above
#have now been censored at the end of their first row of data
dim(roott2.cens)
```

```
[1] 2982    17
```

```
#Fit Cox model to the modified data
cox.adj<-coxph(Surv(stop_time,outcome)~hormon+age+meno+size+
               as.factor(grade)+enodes+pgr+er+year_cat+chemo,
               data=roott2.cens,x=TRUE)
summary(cox.adj)
```

Call:

```
coxph(formula = Surv(stop_time, outcome) ~ hormon + age + meno +
      size + as.factor(grade) + enodes + pgr + er + year_cat +
      chemo, data = roott2.cens, x = TRUE)
```

n= 2982, number of events= 1205

	coef	exp(coef)	se(coef)	z	Pr(> z)
hormon	-0.3202358	0.7259778	0.0959388	-3.338	0.000844 ***

age	-0.0041405	0.9958680	0.0038852	-1.066	0.286549	
meno	0.0254657	1.0257927	0.1022832	0.249	0.803382	
size>50	0.5584043	1.7478811	0.0975290	5.726	1.03e-08	***
size20-50	0.3468098	1.4145477	0.0671587	5.164	2.42e-07	***
as.factor(grade)3	0.3654311	1.4411351	0.0745880	4.899	9.62e-07	***
enodes	-1.9411143	0.1435439	0.1111849	-17.458	< 2e-16	***
pgr	-0.0002233	0.9997768	0.0001258	-1.774	0.076046	.
er	-0.0002042	0.9997958	0.0001213	-1.683	0.092312	.
year_cat	-0.1588427	0.8531305	0.0679124	-2.339	0.019339	*
chemo	-0.3621450	0.6961814	0.0836440	-4.330	1.49e-05	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

	exp(coef)	exp(-coef)	lower .95	upper .95
hormon	0.7260	1.3775	0.6015	0.8762
age	0.9959	1.0041	0.9883	1.0035
meno	1.0258	0.9749	0.8395	1.2535
size>50	1.7479	0.5721	1.4438	2.1161
size20-50	1.4145	0.7069	1.2401	1.6136
as.factor(grade)3	1.4411	0.6939	1.2451	1.6680
enodes	0.1435	6.9665	0.1154	0.1785
pgr	0.9998	1.0002	0.9995	1.0000
er	0.9998	1.0002	0.9996	1.0000
year_cat	0.8531	1.1722	0.7468	0.9746
chemo	0.6962	1.4364	0.5909	0.8202

Concordance= 0.69 (se = 0.008)

Likelihood ratio test= 529.3 on 11 df, p=<2e-16

Wald test = 602.3 on 11 df, p=<2e-16

Score (logrank) test = 668.9 on 11 df, p=<2e-16

- The artificial censoring means cutting off a person's follow-up when they switch from `hormon=0` to `hormon=1`. Here the artificial censoring is implemented simply by removing the second row of data for the 264 switchers. So we are left with 2982 rows again (one row for each individual). Everyone in the modified data is following one of the two treatment strategies of interest from time 0: (i) Start hormone therapy directly following surgery (and continue), (ii) Do not start hormone therapy directly following surgery or for the duration of follow-up.
- The HR estimate for hormone therapy is 0.73 (95% CI 0.60 to 0.88). Conditional on the other variables in the model, treatment strategy (i) is associated with a 27% lowering of the hazard for the composite outcome of disease occurrence or death, compared with treatment strategy (ii). The 95% CI excludes 1, and there is strong evidence against the null hypothesis of no effect of hormone therapy use on the outcome.

Question 3

- What has been assumed about censoring in the Cox model fitted in Question 2?
- Under what assumptions can the HR for hormone therapy be interpreted as the causal effect of treatment strategy (i) versus (ii) on the outcome.
- It is assumed that the censoring is conditionally independent, given the covariates included in the model.
- The HR for hormone therapy can be interpreted as the causal effect of treatment strategy (i) versus treatment strategy (ii) if:
 - (a) switching from `hormon=0` to `hormon=1` depends only on the set of covariates included in the model,
 - (b) the covariates in the model include all confounders of the association between hormone therapy status at time 0 and the outcome,
 - (c) the model is correctly specified.

Question 4

- Using the adjusted Cox model from question 2, obtain an estimate of the risk of recurrence or death up to 5 years for each individual in the data under the two strategies of interest: (i) Start hormone therapy directly following surgery (and continue), (ii) Do not start hormone therapy directly following surgery or for the duration of follow-up.
- For person 1, what would the estimated risk of recurrence or death up to 5 years have been if they had followed treatment strategies (i) and (ii)?

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```
#create dataset in which hormon is set to 1 for everyone
data.hormon1<-roth2.cens[roth2.cens$rownum==1,]
data.hormon1$hormon<-1

#create dataset in which hormon is set to 0 for everyone
data.hormon0<-roth2.cens[roth2.cens$rownum==1,]
data.hormon0$hormon<-0

#predicted risk up to time 5 for each individual
risk.hormon1.time5<-1-predictSurvProb(cox.adj,newdata=data.hormon1,times=5)
risk.hormon0.time5<-1-predictSurvProb(cox.adj,newdata=data.hormon0,times=5)
```

```
#Take a look at the estimates for the first 5 people
cbind(risk.hormon1.time5,risk.hormon0.time5)[1:5,]
```

```
      [,1]      [,2]
[1,] 0.2010685 0.2659739
[2,] 0.2892535 0.3751956
[3,] 0.1886118 0.2501630
[4,] 0.3321864 0.4265825
[5,] 0.2159383 0.2847260
```

For an individual with the same characteristics as person 1 in the data, except for their hormone therapy status, the risk of recurrence or death up to 5 years is estimated to be 20% if they were to take hormone therapy and 27% if they were not to take hormone therapy.

Question 5

- Using your model from Question 4, obtain estimates of the marginal risks up to time 5, $\Pr(T \leq 5|do(X = 1))$ and $\Pr(T \leq 5|do(X = 0))$, where $X = 1$ denotes treatment strategy (i) and $X = 0$ denotes treatment strategy (ii).
- Obtain estimates of the marginal risk difference and marginal risk ratio and interpret your results.

You can do this ‘by hand’ using the conditional survival probabilities obtained in Question 4, or using `adjustedsurv`.

SOLUTIONS

```
#---
#mean risk at each time, averaging over all individuals
riskmean.hormon1.time5<-mean(risk.hormon1.time5)
riskmean.hormon0.time5<-mean(risk.hormon0.time5)

#marginal risk diff and risk ratio
risk.diff<-(riskmean.hormon1.time5)-(riskmean.hormon0.time5)
risk.ratio<-(riskmean.hormon1.time5)/(riskmean.hormon0.time5)

riskmean.hormon1.time5
```

```
[1] 0.3634107
```

```
riskmean.hormon0.time5
```

```
[1] 0.4487472
```

```
risk.diff
```

```
[1] -0.08533657
```

```
risk.ratio
```

```
[1] 0.8098338
```

```
#---  
#same thing using adjustedCurves, with approx confidence intervals  
  
#Refit Cox model with hormon as factor  
roth2.cens$hormonfac<-as.factor(roth2.cens$hormon)  
cox.adjfac<-coxph(Surv(stop_time,outcome)~hormonfac+age+meno+size+  
                  as.factor(grade)+enodes+pgr+er+year_cat+chemo,  
                  data=roth2.cens,x=TRUE)  
  
adjsurv <- adjustedsurv(data=roth2.cens,  
                        variable="hormonfac",  
                        ev_time="stop_time",  
                        event="outcome",  
                        method="direct",  
                        outcome_model=cox.adjfac,  
                        conf_int=TRUE,  
                        bootstrap=F,times=5)  
  
#risks at time 5  
adjsurv$ate_object$meanRisk
```

	estimator	time	treatment	estimate	se	lower	upper
	<char>	<num>	<fctr>	<num>	<num>	<num>	<num>
1:	GFORMULA	5	0	0.4487472	0.01006017	0.4290297	0.4684648
2:	GFORMULA	5	1	0.3634107	0.02130564	0.3216524	0.4051690

```
#risk diff at time 5
adjsurv$aate_object$diffRisk
```

```

      estimator  time      A estimate.A      B estimate.B      estimate      se
      <char> <num> <char>      <num> <char>      <num>      <num>
1:  GFORMULA    5      0  0.4487472    1  0.3634107 -0.08533657  0.02298069
      lower      upper      p.value
      <num>      <num>      <num>
1: -0.1303779 -0.04029525  0.0002044903

```

```
#risk ratio at time 5
adjsurv$aate_object$ratioRisk
```

```

      estimator  time      A estimate.A      B estimate.B      estimate      se
      <char> <num> <char>      <num> <char>      <num>      <num>
1:  GFORMULA    5      0  0.4487472    1  0.3634107  0.8098338  0.04975028
      lower      upper      p.value
      <num>      <num>      <num>
1:  0.712325  0.9073425  0.0001321507

```

The estimates are shown in the table below. All results are focused on a comparison of marginal risks under the treatment strategies of: (i) Start hormone therapy following surgery (and continue as recommended by the doctor), (ii) Do not start hormone therapy following surgery or for the duration of follow-up of interest.

Estimand	Estimate	95% CI
Marginal risk up to 5 years: hormon=1	0.36	0.32 to 0.41
Marginal risk up to 5 years: hormon=0	0.45	0.43 to 0.47
Marginal risk difference	-0.085	-0.130 to -0.040
Marginal risk ratio	0.81	0.71 to 0.91

```
#-----
#Not asked for in the practical,
#but in case you also want to get the full survival curves...

adjsurv <- adjustedsurv(data=rotd2.cens,
                        variable="hormonfac",
                        ev_time="stop_time",
                        event="outcome",
```

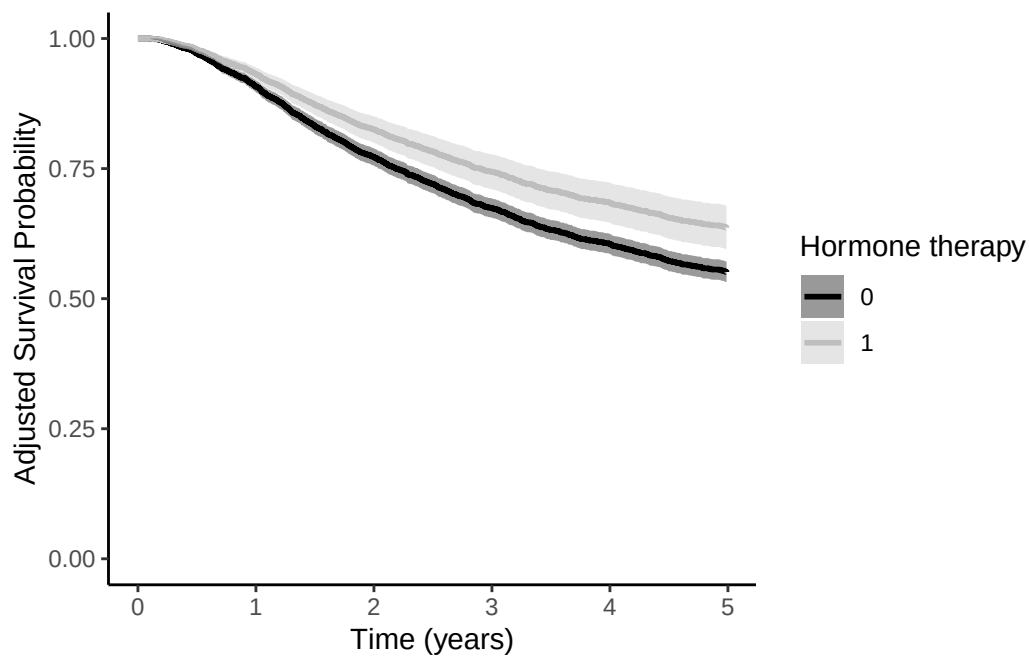
```

method="direct",
outcome_model=cox.adjfac,
conf_int=TRUE,
bootstrap=F)

plot(adjsurv,conf_int=T,custom_colors=c("black","grey"),
     xlab="Time (years)",
     legend.title="Hormone therapy",ylim=c(0,1))

```

Loading required namespace: pammttools



Part B: Competing events

In this part return to using the data from Practical 10 (`rotterdam_p10.csv`), in which treatment status is fixed over time. We now consider the outcomes of ‘disease recurrence’ and ‘death without recurrence’ as competing events. The variables were summarised in the table at the start of Practical 10.

Question 1

- Load the data. The variable `time` is years to the first of recurrence or death without recurrence. The variable `outcome_cr` indicates what happened to the person at the end of their follow-up: 0=censored, 1=recurrence, 2=death without recurrence
- How many people had each of the two competing events?
- Obtain a plot of the Aalen-Johansen estimates of the cumulative incidences of the two event types by hormone therapy use. You can look back to Practical 6 (Question 6) for code to do this.

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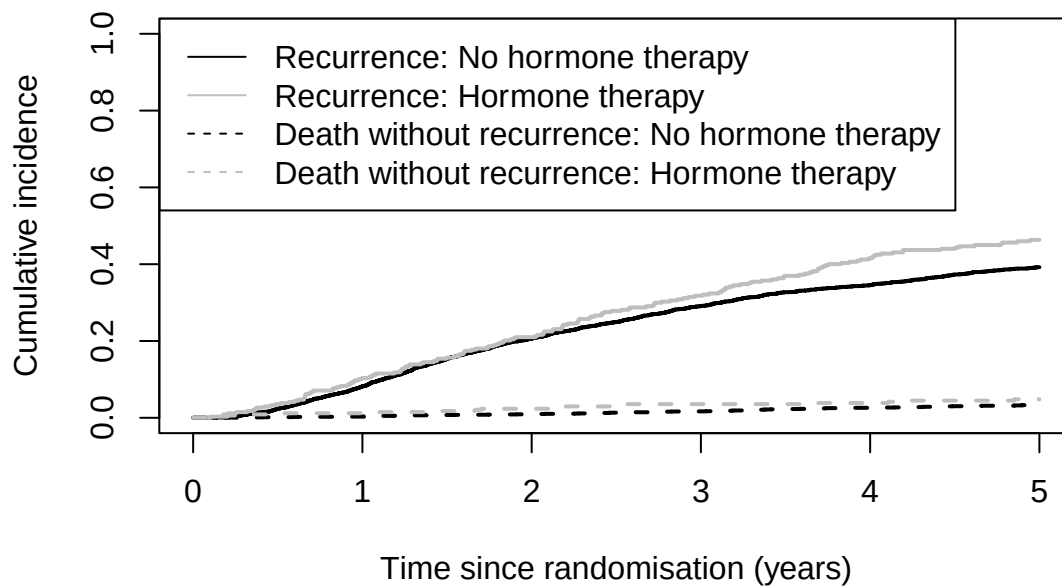
```
#load the data
rottp<-read.table("datasets/rotterdam_p10.csv",sep="," ,header=T)

table(rottp$outcome_cr)
```

```
      0      1      2
1697 1181  104
```

```
#Cumulative incidence plot
cinc.hormon <- survfit(Surv(time, as.factor(outcome_cr)) ~ hormon,
                      data=rottp)

plot(cinc.hormon,lwd=2,
     lty=c(1,1,2,2),col=c("black","grey","black","grey"),
     xlab="Time since randomisation (years)",
     ylab="Cumulative incidence",
     ylim=c(0,1))
legend("topleft",c("Recurrence: No hormone therapy",
                  "Recurrence: Hormone therapy",
                  "Death without recurrence: No hormone therapy",
                  "Death without recurrence: Hormone therapy"),
     col=c("black","grey","black","grey"),
     lty=c(1,1,2,2))
```



- 1181 individuals had disease recurrence and 104 died without disease recurrence. The remaining 1697 were censored.
- The plot shows that hormone therapy users had higher incidence of both recurrence and death without recurrence.

Question 2

In this question we will estimate conditional cumulative incidences of the two event types under hormone therapy use and non-use for each individual in the data. Use the code below to do the following:

- Fit cause-specific Cox models for each of the two event types of interest, including hormone therapy as the main exposure and adjusting for the same set of variables used above.
- Use `survfit` to obtain estimates of the cumulative incidences for the two event types for each individual under the two treatment strategies (setting `hormon=1` and `hormon=0`).
- Take the means across individuals to obtain the marginal cumulative incidences under the two treatment strategies.

Interpret the resulting plot.

```
#---
#Fit cause-specific Cox models
cox.adj.cr<-coxph(Surv(time,as.factor(outcome_cr))~hormon+age+meno+
                  size+as.factor(grade)+enodes+pgr+er+year_cat+chemo,
                  data=rott,id=pid)
```

```

#---
#Conditional cumulative incidences under each treatment strategy

#create dataset in which hormon is set to 1 for everyone
data.hormon1<-roth
data.hormon1$hormon<-1

#create dataset in which hormon is set to 0 for everyone
data.hormon0<-roth
data.hormon0$hormon<-0

#obtain cumulative incidences for each individual given their covariates
#under each treatment strategy
coxpred.1 <- survfit(cox.adj.cr, newdata= data.hormon1)
coxpred.0 <- survfit(cox.adj.cr, newdata= data.hormon0)

#cumulative incidence for recurrence (pstate[,2])
#under each treatment strategy
#at all event times
cinc.recur.1.individual<-coxpred.1$pstate[,2]
cinc.recur.0.individual<-coxpred.0$pstate[,2]

#cumulative incidence for death without recurrence (pstate[,3])
#under each treatment strategy
#at all event times
cinc.death.1.individual<-coxpred.1$pstate[,3]
cinc.death.0.individual<-coxpred.0$pstate[,3]

#---
#Marginal cumulative incidences under each treatment strategy
#at each event time

cinc.recur.1<-rowMeans(cinc.recur.1.individual)
cinc.recur.0<-rowMeans(cinc.recur.0.individual)

cinc.death.1<-rowMeans(cinc.death.1.individual)
cinc.death.0<-rowMeans(cinc.death.0.individual)

#---
#plot marginal cumulative incidence curves

times<-coxpred.1$time#list of all event times

```

```

plot(times,cinc.recur.0,type="s",col="black",lty=1,lwd=2,
      xlim=c(0,5),ylim=c(0,1),
      xlab="Time (years)", ylab="Cumulative incidence")
lines(times,cinc.recur.1,type="s",col="grey",lty=1,lwd=2)
lines(times,cinc.death.0,type="s",col="black",lty=2,lwd=2)
lines(times,cinc.death.1,type="s",col="grey",lty=2,lwd=2)
legend("topleft", c("Recurrence: No hormone therapy",
                    "Recurrence: Hormone therapy",
                    "Death without recurrence: No hormone therapy",
                    "Death without recurrence: Hormone therapy"),
      col=c("black","grey","black","grey"),
      lty=c(1,1,2,2), lwd=2, bty='n')

```

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```

#---
#Fit cause-specific Cox models
cox.adj.cr<-coxph(Surv(time,as.factor(outcome_cr))~hormon+age+meno+
                  size+as.factor(grade)+enodes+pgr+er+year_cat+chemo,
                  data=rott,id=pid)
cox.adj.cr

```

Call:

```

coxph(formula = Surv(time, as.factor(outcome_cr)) ~ hormon +
      age + meno + size + as.factor(grade) + enodes + pgr + er +
      year_cat + chemo, data = rott, id = pid)

```

1:2	coef	exp(coef)	se(coef)	robust se	z
hormon	-0.2882098	0.7496043	0.0981144	0.1014676	-2.840
age	-0.0165860	0.9835508	0.0039311	0.0041744	-3.973
meno	0.1507104	1.1626598	0.1024553	0.1069414	1.409
size>50	0.5841268	1.7934242	0.0982881	0.1041772	5.607
size20-50	0.3747735	1.4546619	0.0681239	0.0685925	5.464
as.factor(grade)3	0.3759814	1.4564200	0.0757840	0.0750864	5.007
enodes	-1.9895531	0.1367565	0.1099589	0.1222438	-16.275
pgr	-0.0003017	0.9996983	0.0001330	0.0001522	-1.983
er	-0.0002056	0.9997944	0.0001272	0.0001358	-1.514
year_cat	-0.1772065	0.8376068	0.0671144	0.0676985	-2.618

```

chemo                -0.3514082  0.7036965  0.0821351  0.0844820  -4.160

1:2                  p
hormon                0.00451
age                   7.09e-05
meno                  0.15875
size>50               2.06e-08
size20-50             4.66e-08
as.factor(grade)3     5.52e-07
enodes                < 2e-16
pgr                   0.04737
er                    0.13013
year_cat              0.00886
chemo                 3.19e-05

1:3                  coef exp(coef)  se(coef)  robust se      z      p
hormon             -0.3994277  0.6707038  0.3251809  0.3077898 -1.298 0.19438
age                 0.1233333  1.1312614  0.0134929  0.0140919  8.752 < 2e-
16
meno               -0.3964329  0.6727154  0.5324086  0.5628075 -0.704 0.48119
size>50             0.3232497  1.3816103  0.3085270  0.3122765  1.035 0.30060
size20-50           0.0277141  1.0281017  0.2300185  0.2370230  0.117 0.90692
as.factor(grade)3 -0.0206468  0.9795649  0.2248881  0.2241917 -0.092 0.92662
enodes              -1.1738302  0.3091805  0.4123264  0.3994055 -2.939 0.00329
pgr                 0.0003502  1.0003502  0.0003145  0.0002218  1.579 0.11439
er                  -0.0006198  0.9993804  0.0003727  0.0003120 -1.986 0.04699
year_cat            -0.0118344  0.9882353  0.2433288  0.2339005 -0.051 0.95965
chemo               -0.4665655  0.6271525  0.6577501  0.6660256 -0.701 0.48360

```

States: 1= (s0), 2= 1, 3= 2

Likelihood ratio test=777.8 on 22 df, p=< 2.2e-16
n= 2982, number of events= 1285

```

#---
#Conditional cumulative incidences under each treatment strategy

#create dataset in which hormon is set to 1 for everyone
data.hormon1<-roth
data.hormon1$hormon<-1

```

```

#create dataset in which hormon is set to 0 for everyone
data.hormon0<-roth
data.hormon0$hormon<-0

#obtain cumulative incidences for each individual given their covariates
#under each treatment strategy
coxpred.1 <- survfit(cox.adj.cr, newdata= data.hormon1)
coxpred.0 <- survfit(cox.adj.cr, newdata= data.hormon0)

#cumulative incidence for recurrence (pstate[,2])
#under each treatment strategy
#at all event times
cinc.recur.1.individual<-coxpred.1$pstate[,2]
cinc.recur.0.individual<-coxpred.0$pstate[,2]

#cumulative incidence for death without recurrence (pstate[,3])
#under each treatment strategy
#at all event times
cinc.death.1.individual<-coxpred.1$pstate[,3]
cinc.death.0.individual<-coxpred.0$pstate[,3]

#---
#Marginal cumulative incidences under each treatment strategy
#at each event time

cinc.recur.1<-rowMeans(cinc.recur.1.individual)
cinc.recur.0<-rowMeans(cinc.recur.0.individual)

cinc.death.1<-rowMeans(cinc.death.1.individual)
cinc.death.0<-rowMeans(cinc.death.0.individual)

#---
#plot marginal cumulative incidence curves

times<-coxpred.1$time#list of all event times

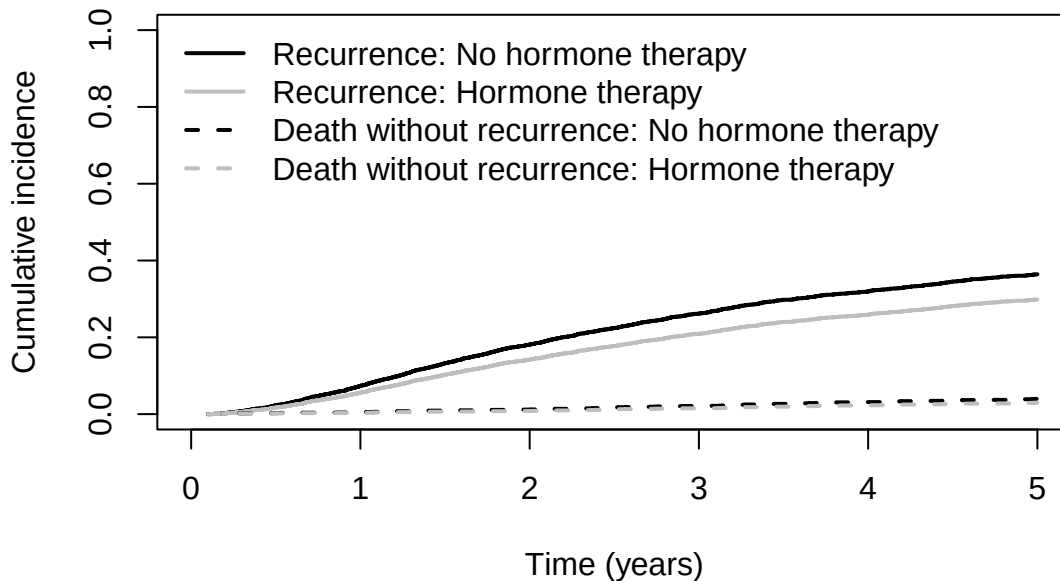
plot(times,cinc.recur.0,type="s",col="black",lty=1,lwd=2,
      xlim=c(0,5),ylim=c(0,1),
      xlab="Time (years)", ylab="Cumulative incidence")
lines(times,cinc.recur.1,type="s",col="grey",lty=1,lwd=2)
lines(times,cinc.death.0,type="s",col="black",lty=2,lwd=2)
lines(times,cinc.death.1,type="s",col="grey",lty=2,lwd=2)

```

```

legend("topleft", c("Recurrence: No hormone therapy",
                    "Recurrence: Hormone therapy",
                    "Death without recurrence: No hormone therapy",
                    "Death without recurrence: Hormone therapy"),
      col=c("black","grey","black","grey"),
      lty=c(1,1,2,2), lwd=2, bty='n')

```



- In the cause-specific hazard model for disease recurrence the HR for hormone therapy is 0.750 (95% CI 0.619 to 0.909), after conditioning on the other variables. That is the hazard of disease recurrence is estimated to be lower by 25% with hormone therapy use versus non-use, conditional on the other variables, and this is statistically significantly different from 1.
- In the cause-specific hazard model for death without recurrence the HR for hormone therapy is 0.671 (95% CI 0.355 to 1.268), after conditioning on the other variables. That is, the hazard of death without recurrence is estimated to be lower by 33% with hormone therapy use versus non-use, but the confidence interval is wide and includes 1, meaning there is no evidence of an association between hormone therapy use and the hazard for death without recurrence conditional on the other variables.
- After adjusting for the identified set of potential confounders, use of hormone therapy is estimated to lower the risk of both disease recurrence and (to a lesser extent) death without recurrence.
- Marginal estimates interpretation: If everyone were to receive hormone therapy the risk (cumulative incidence) of disease recurrence and death would have been lower than if everyone had not received hormone therapy.

Question 3

The marginal cumulative incidences can also be obtained in the `adjustedCurves` package using `adjustedcif`. Take a look at the code below for doing this.

Note this this requires the cause-specific Cox models to be fitted using CSC in the `riskRegression` package.

```
#---
#Fit cause-specific hazard models using CSC from the package riskRegression
#`hormon` needs to be a factor variable
roth$hormonfac<-as.factor(roth$hormon)
cox.adj.cr2<-CSC(
  Hist(time, outcome_cr)~hormonfac+age+meno+size+as.factor(grade)+
  enodes+pgr+er+year_cat+chemo,data=roth)

#---
#Now use adjustedcif to get the standardised cumulative incidence curves
#This has to be done separately for each event type

#recurrence
adjcif1 <- adjustedcif(data=roth,
                      variable="hormonfac",
                      ev_time="time",
                      event="outcome_cr",
                      method="direct",
                      outcome_model=cox.adj.cr2,
                      cause=1,
                      conf_int=TRUE)
plot(adjcif1, conf_int=TRUE,custom_colors=c("black","grey"),
     xlab="Time (years)",legend.title="Hormone therapy",
     ylim=c(0,0.5),title = "Disease recurrence")

#Death without recurrence
adjcif2 <- adjustedcif(data=roth,
                      variable="hormonfac",
                      ev_time="time",
                      event="outcome_cr",
                      method="direct",
                      outcome_model=cox.adj.cr2,
                      cause=2,
                      conf_int=TRUE)
plot(adjcif2, conf_int=TRUE,custom_colors=c("black","grey"),
```

```
xlab="Time (years)",legend.title="Hormone therapy",
ylim=c(0,0.05),title = "Death without recurrence")
```

We can also use `adjustedcif` to obtain cumulative incidence estimates at a specific time, as well as risk differences and risk ratios. This is shown below for time 5.

```
#Recurrence
adjcif1.time5 <- adjustedcif(data=roth,
                             variable="hormonfac",
                             ev_time="time",
                             event="outcome_cr",
                             method="direct",
                             outcome_model=cox.adj.cr2,
                             cause=1,
                             conf_int=TRUE,
                             times = 5)

adjcif1.time5$ate_object$meanRisk
adjcif1.time5$ate_object$diffRisk
adjcif1.time5$ate_object$ratioRisk
```

```
#Death without recurrence
adjcif2.time5 <- adjustedcif(data=roth,
                             variable="hormonfac",
                             ev_time="time",
                             event="outcome_cr",
                             method="direct",
                             outcome_model=cox.adj.cr2,
                             cause=2,
                             conf_int=TRUE,
                             times = 5)

adjcif2.time5$ate_object$meanRisk
adjcif2.time5$ate_object$diffRisk
adjcif2.time5$ate_object$ratioRisk
```

SOLUTIONS

```
#---
#Fit cause-specific hazard models using CSC from the package riskRegression
#`hormon` needs to be a factor variable
```

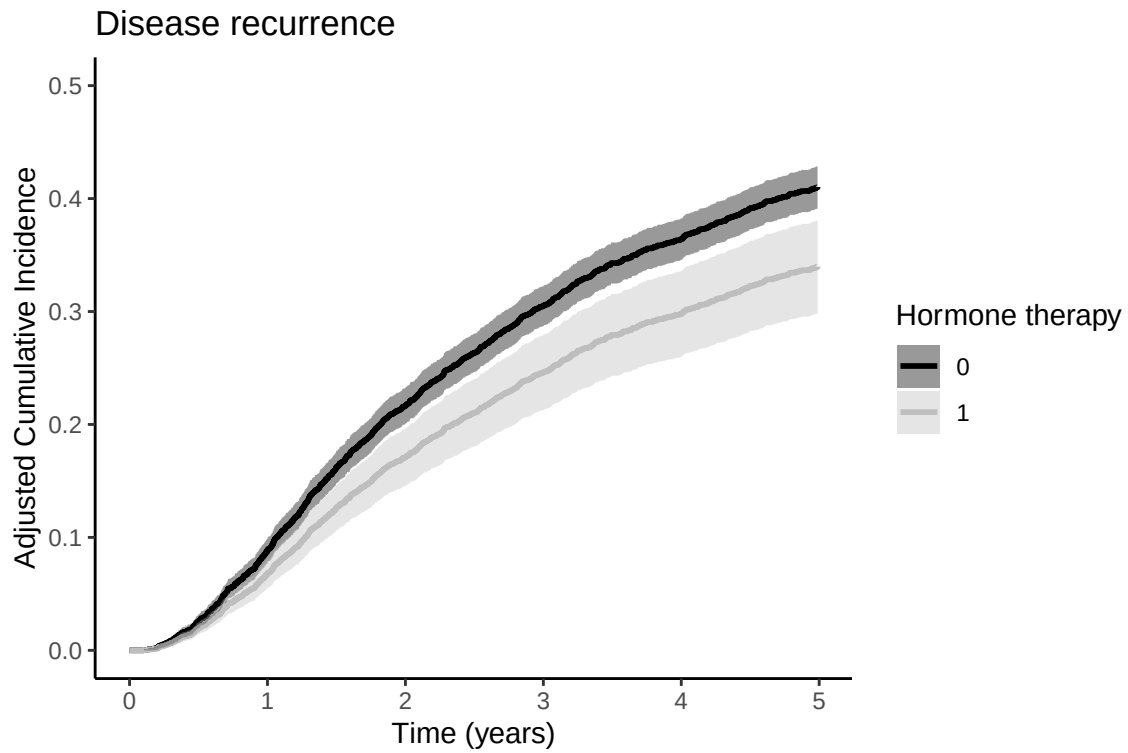
```

rottp$hormonfac<-as.factor(rottp$hormon)
cox.adj.cr2<-CSC(
  Hist(time, outcome_cr)~hormonfac+age+meno+size+as.factor(grade)+
    enodes+pgr+er+year_cat+chemo,data=rottp)

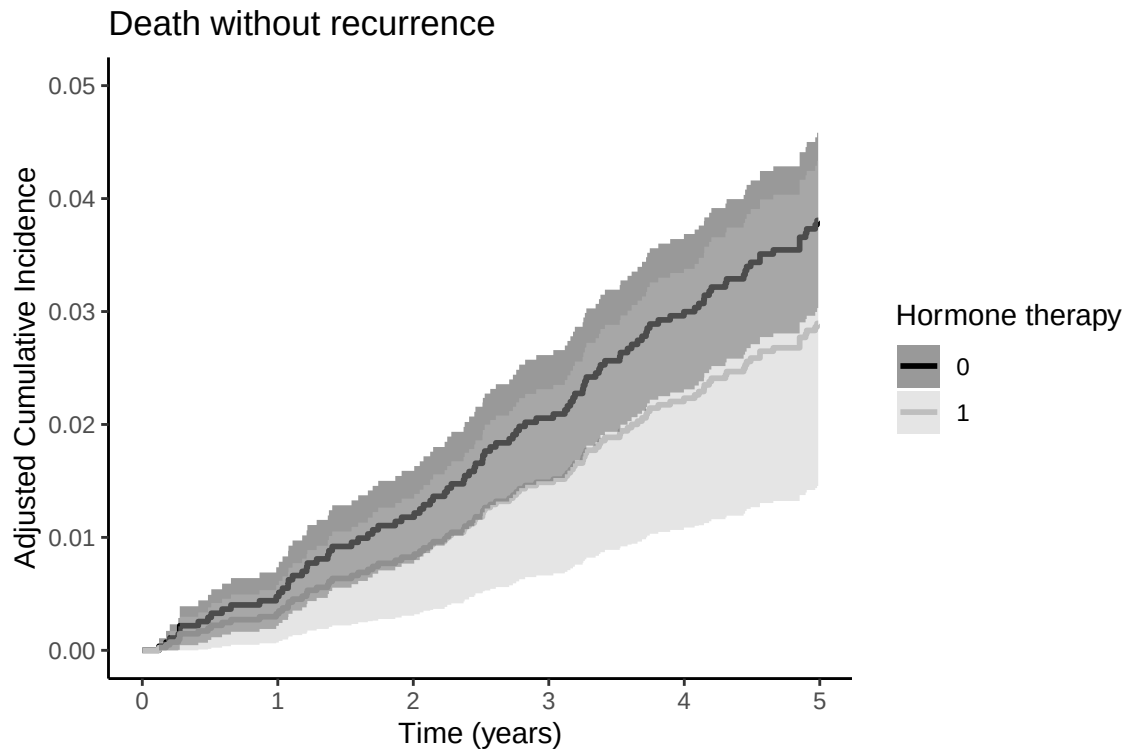
#---
#Now use adjustedcif to get the standardised cumulative incidence curves
#This has to be done separately for each event type

#recurrence
adjcif1 <- adjustedcif(data=rottp,
  variable="hormonfac",
  ev_time="time",
  event="outcome_cr",
  method="direct",
  outcome_model=cox.adj.cr2,
  cause=1,
  conf_int=TRUE)
plot(adjcif1, conf_int=TRUE,custom_colors=c("black","grey"),
  xlab="Time (years)",legend.title="Hormone therapy",
  ylim=c(0,0.5),title = "Disease recurrence")

```



```
#Death without recurrence
adjcif2 <- adjustedcif(data=rott,
                      variable="hormonfac",
                      ev_time="time",
                      event="outcome_cr",
                      method="direct",
                      outcome_model=cox.adj.cr2,
                      cause=2,
                      conf_int=TRUE)
plot(adjcif2, conf_int=TRUE, custom_colors=c("black", "grey"),
     xlab="Time (years)", legend.title="Hormone therapy",
     ylim=c(0, 0.05), title = "Death without recurrence")
```



```
#Recurrence
adjcif1.time5 <- adjustedcif(data=rott,
                             variable="hormonfac",
                             ev_time="time",
                             event="outcome_cr",
                             method="direct",
                             outcome_model=cox.adj.cr2,
                             cause=1,
                             conf_int=TRUE,
                             times = 5)
adjcif1.time5$ate_object$meanRisk
```

	estimator	time	treatment	estimate	se	lower	upper
	<char>	<num>	<fctr>	<num>	<num>	<num>	<num>
1:	GFORMULA	5	0	0.4101759	0.009564185	0.3914305	0.4289214
2:	GFORMULA	5	1	0.3397203	0.021071665	0.2984206	0.3810200

```
adjcif1.time5$ate_object$diffRisk
```

estimator	time	A estimate.A	B estimate.B	estimate	se
-----------	------	--------------	--------------	----------	----

	<char>	<num>	<char>	<num>	<char>	<num>	<num>	<num>
1:	GFORMULA	5	0	0.4101759	1	0.3397203	-0.07045562	0.02255436
	lower		upper	p.value				
	<num>		<num>	<num>				
1:		-0.1146614	-0.02624989	0.001785232				

```
adjcif1.time5$ate_object$ratioRisk
```

	estimator	time	A	estimate.A	B	estimate.B	estimate	se
	<char>	<num>	<char>	<num>	<char>	<num>	<num>	<num>
1:	GFORMULA	5	0	0.4101759	1	0.3397203	0.8282307	0.05366736
	lower		upper	p.value				
	<num>		<num>	<num>				
1:		0.7230446	0.9334168	0.001371284				

```
#Death without recurrence
adjcif2.time5 <- adjustedcif(data=roth,
                             variable="hormonfac",
                             ev_time="time",
                             event="outcome_cr",
                             method="direct",
                             outcome_model=cox.adj.cr2,
                             cause=2,
                             conf_int=TRUE,
                             times = 5)
adjcif2.time5$ate_object$meanRisk
```

	estimator	time	treatment	estimate	se	lower	upper
	<char>	<num>	<fctr>	<num>	<num>	<num>	<num>
1:	GFORMULA	5	0	0.03806647	0.003960401	0.03030422	0.04582871
2:	GFORMULA	5	1	0.02894646	0.007346797	0.01454701	0.04334592

```
adjcif2.time5$ate_object$diffRisk
```

	estimator	time	A	estimate.A	B	estimate.B	estimate	se
	<char>	<num>	<char>	<num>	<char>	<num>	<num>	<num>
1:	GFORMULA	5	0	0.03806647	1	0.02894646	-0.009120003	0.008528223
	lower		upper	p.value				
	<num>		<num>	<num>				
1:		-0.02583501	0.007595007	0.2848936				

```
adjcif2.time5$ate_object$ratioRisk
```

```

      estimator  time      A estimate.A      B estimate.B estimate      se
      <char> <num> <char>      <num> <char>      <num>      <num>      <num>
1: GFORMULA      5      0 0.03806647      1 0.02894646 0.760419 0.212412
      lower      upper    p.value
      <num>      <num>      <num>
1: 0.3440992 1.176739 0.2593591

```

The estimates are shown in the table below. The 95% CIs have been obtained using an approximation. If everyone had received hormone therapy we would expected to have seen lower risk of recurrence compared with if no one had received hormone therapy. For example the marginal cumulative incidence of recurrence is estimated to be reduced by 17% (95% CI 7% to 28%) if everyone had taken hormone therapy versus not. However, there is no evidence of an effect of hormone therapy use on the incidence of death without recurrence up to 5 years.

Estimand	Estimate	95% CI
Recurrence		
Marginal cumulative incidence (risk) up to 5 years: hormon=1	0.34	0.29 to 0.38
Marginal cumulative incidence (risk) up to 5 years: hormon=0	0.41	0.39 to 0.43
Marginal risk difference	-0.070	-0.115 to -0.026
Marginal risk ratio	0.83	0.72 to 0.93
Death without recurrence		
Marginal cumulative incidence (risk) up to 5 years: hormon=1	0.029	0.014 to 0.043
Marginal cumulative incidence (risk) up to 5 years: hormon=0	0.038	0.030 to 0.045
Marginal risk difference	-0.009	-0.026 to 0.008
Marginal risk ratio	0.76	0.34 to 1.18